Flex-Act: Why Learn when you can Pick?

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Abstract

Learning activation functions has emerged as a promising direction in deep learning, allowing networks to adapt activation mechanisms to task-specific demands. In this work, we introduce a novel framework that employs the Gumbel-Softmax trick to enable discrete yet differentiable selection among a predefined set of activation functions during training. Our method dynamically learns the optimal activation function independently of the input, thereby enhancing both predictive accuracy and architectural flexibility. Experiments on synthetic datasets show that our model consistently selects the most suitable activation function, underscoring its effectiveness. These results connect theoretical advances with practical utility, paving the way for more adaptive and modular neural architectures in complex learning scenarios.

1 Introduction

While modern deep learning architectures have revolutionized domains spanning vision, language, and control, a surprisingly under-explored axis in this progression lies in the choice and design of activation functions. These pointwise non-linearities not only endow neural networks with the capacity to model complex functions, but also profoundly influence trainability, signal propagation, and generalization behavior (Nair & Hinton, 2010; Nwankpa et al., 2018; Pedamonti, 2018; Subramanian et al., 2024). As we scale models to unprecedented depths and widths, activation functions become increasingly critical mediators of inductive bias, information flow, and learning dynamics. Yet, in most empirical pipelines, the activation function is treated as a fixed hyperparameter—often defaulting to ReLU, GELU, or their minor variants—without sufficient scrutiny of their behavior under scaling regimes. Recent work on scaling laws (Kaplan et al., 2020) has identified predictable trends in model performance with respect to compute, data, and parameter count. These observations have shaped the design of large-scale transformers and vision architectures, spurring innovations in initialization (Saxe et al., 2014; He et al., 2015a), normalization (Ba et al., 2016; Zhang et al., 2019), and architecture (Vaswani, 2017; Dosovitskiy et al., 2021). However, activation functions remain a surprisingly static component within this dynamic progress of neural networks, and their ongoing research. When scaled to hundreds of layers, signal fidelity and gradient flow hinges on the choice of non-linearity: whether activations preserve variance across layers (Glorot & Bengio, 2010), whether they induce saturation or sparsity (Maas et al., 2013), and whether their derivatives yield favorable Hessian spectra (Pennington et al., 2017) — all of these directions emerge as pivotal factors in the success of large models.

From the perspective of representational power, activation functions define the functional class accessible to the network. A poor choice can lead to shattered gradients, and unstable dynamics (Balduzzi et al., 2018). For instance, the introduction of Swish and GELU brought improved performance through smoother transitions and better gradient characteristics (Ramachandran et al., 2017; Hendrycks & Gimpel, 2016), yet these improvements were often empirical and lacked a deep mechanistic understanding in high-dimensional, deep neural networks. Crucially, in the context of pretraining and finetuning large models, activation functions also affect the scaling of logits and their calibration, with downstream effects on optimization dynamics, learning rate schedules, and generalization (Zhang et al., 2017). Moreover, new paradigms such as retrieval-augmented generation, multi-modal modeling, and continuous pretraining demand activations that are not only expressive and efficient, but also compositional and stable across wide operational regimes.

Despite these considerations, our understanding of how activation functions interact with the deep geometry of networks remains primitive. Unlike advances in architecture search or self-attention mechanisms, there is a dearth of systematic studies that investigate the optimality of activations in the infinite-width, deep-scaling, or heavy-tailed spectrum regimes. Activation functions are the crux of information transformation in neural architectures and it is increasingly evident that scaling neural networks without revisiting their nonlinear cores is a missed opportunity. In this work, we argue for a re-examination of activation function design in the context of deep scaling. We unify insights from theoretical neuroscience, kernel regimes, and empirical large-model behavior to highlight the crucial yet neglected role activations play in efficient, expressive, and stable learning at scale. We motivate this inquiry through the lens of dynamical systems, signal propagation theory, and spectral conditioning, setting the stage for a principled exploration of scalable non-linearities. The widespread success of deep learning hinges on scalable architectures whose expressiveness is governed not only by depth and width, but also by the choice of activation functions. Conventional activations such as ReLU (Nair & Hinton, 2010), GELU (Hendrycks & Gimpel, 2016), and Swish (Ramachandran et al., 2017) are statically defined, applied identically across layers, tasks, and data regimes. However, this static design paradigm limits adaptability and ignores the growing evidence that no single non-linearity is universally optimal across depth or domain. This mismatch becomes especially pronounced in deep networks where representational demands evolve layer-wise.

Motivation. Standard neural networks commit to a fixed activation function before training begins. While some functions generalize well across tasks, no single activation performs optimally across the spectrum of functional regimes encountered in practice. This inflexibility becomes particularly limiting when one wishes to deploy the *same architecture* across multiple datasets or target mappings, where the most effective non-linearity may differ. For example, a task involving bounded, saturating behavior may favor a Sigmoid or Tanh, while another may benefit from the sparsity and unboundedness of ReLU. Recent works have proposed parameterized or learnable activations that morph during training (He et al., 2015b; Tavakoli et al., 2020), or even architectures that learn the entire activation function as a neural approximator (Liu et al., 2024). While expressive, these methods often come with higher optimization complexity, reduced interpretability, and the need to commit to a specific parameterization class. Instead, we explore a simpler but powerful alternative: allow each layer to *choose* from a fixed basis of standard activation functions. The selection is learned dynamically during training, and is jointly optimized with model weights. This enables a single network—without any manual hyperparameter tuning or function sweeping—to recover the appropriate non-linearity for the task at hand. We demonstrate this in synthetic regression tasks where the ground-truth activation is known but hidden: our model discovers and adopts the correct function without needing to be retrained across tasks.

Overview and Evaluation. We introduce `Flex-Act`, a framework for discrete activation selection based on Gumbel-Softmax routing. Each layer selects its activation from a small set of candidate functions, with selection probabilities learned through backpropagation. To mitigate biases toward unbounded activations, we introduce a novel gradient-norm-based regularizer that aligns routing choices with functional suitability. `Flex-Act` is evaluated in settings where the true activation function governing the target is known, and we compare its performance against fixed activation baselines and parameterized methods. Our results show that `Flex-Act` consistently adapts to the correct activation function without requiring multiple models, activation sweeps, or architectural tuning enabling plug-and-play generalization across diverse function classes. This capability makes it a promising tool for robust, general-purpose deep learning pipelines.

Contributions. This paper introduces a new paradigm for non-linearity design in deep networks through *discrete activation selection*, where each layer dynamically routes its input through one of several candidate activation functions. Our core contributions are as follows:

- **Discrete Activation Routing via Gumbel-Softmax.** We present a novel mechanism for layer-wise selection of activation functions using the Gumbel-Softmax reparameterization trick, allowing discrete function choices to be optimized end-to-end via gradient descent. (see Section 3.2)
- **Gradient Derivations for Discrete Non-Linearities.** We analytically derive the backpropagation equations for networks employing discrete activation routing, overcoming challenges posed by non-differentiable selection. (see Section 3.2)
- **Empirical Gains in Deep Settings.** Our approach consistently improves accuracy and training stability over fixed or parameterized activation baselines, particularly in very deep architectures where functional diversity across layers is critical. (see Section 4)
- **Interpretable and Modular Non-Linearity Design.** By exposing activation selection as a discrete, learnable variable, our framework enables new avenues for analyzing functional usage across depth, offering both interpretability and architectural flexibility. (see Section 5)

2 Related Work

Activation functions are fundamental components of neural networks, providing non-linear transformations that are necessary for approximating complex functions. Their design and choice significantly influence the performance and trainability of deep learning models. This section reviews the evolution of activation function research, from traditional functions to parameterized approaches. We also note there is significant overlap with the literature in online learning and bandits. Throughout this section, we highlight how our work differs by focusing on discrete activation selection tailored to layer-specific requirements. The study of activation functions in neural networks spans from classical hand-crafted forms to recent efforts in adaptive and learnable designs. In this section, we organize prior work into: (i) Traditional Fixed Activations, (ii) Parameterized Activation Functions, and (iii) Discrete or Learnable Function Routing.

Traditional Fixed Activation Functions. Early neural networks relied on activation functions such as Sigmoid and Tanh, which introduced the essential non-linearity required to model complex relationships. These functions were instrumental in enabling neural networks to approximate arbitrary functions. However, as networks grew deeper, their limitations became evident. Both Sigmoid and Tanh functions suffered from the vanishing gradient problem, where gradients become increasingly small as they propagate through layers. This hindered the ability of earlier layers to update effectively during training as discussed in Nair & Hinton (2010). ReLU enabled deeper networks by improving gradient flow and computational efficiency (Nair & Hinton, 2010). However, ReLU itself introduced issues such as the "dying ReLU" problem, where neurons become inactive and cease to contribute to learning if the inputs become negative. Variants such as Leaky ReLU (Maas et al., 2013) and ELU (Exponential Linear Unit) (Clevert, 2015) were proposed to mitigate these issues, allowing small, non-zero gradients for negative inputs and ensuring smoother transitions. Another significant advancement came with GELU (Gaussian Error Linear Unit) (Hendrycks & Gimpel, 2016), a function that combines the strengths of ReLU and Sigmoid. GELU applies a smooth, differentiable approximation of a gating mechanism, blending linear and non-linear behaviors. It has shown particular effectiveness in natural language processing models like BERT (Devlin, 2018) and transformer architectures (Vaswani, 2017). GELU ensures smoother gradients and better convergence properties, making it a popular choice in many state-of-the-art architectures. Despite these innovations, traditional activation functions remain fixed across layers and tasks, limiting their adaptability to diverse requirements. Our work builds on the recognition that a single, fixed activation function may not be optimal for all layers in a deep network. Instead of relying on fixed functions or parameterizing a single family, we propose a framework where layers dynamically select the most appropriate activation function from a predefined set, addressing layer-specific (primarily penultimate